



2025

022

BOOKLET NO.

148117

Geology
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

P.T.O.

**SECTION – A**

- Q1.** Answer the following questions in about **150** words **each**. **(5×10=50)**
- Q1.** (a) Geochemical model of the earth's crust. **10**
- Q1.** (b) Fundamental principles of Geomorphology. **10**
- Q1.** (c) Applications of remote sensing. **10**
- Q1.** (d) Various types of unconformities. **10**
- Q1.** (e) Seismic waves and seismic zones in India. **10**
-
- Q2.** (a) What features are noticed at plate boundaries and discuss briefly on tectonics ? **20**
- Q2.** (b) Discuss nature and origin of solar system. **15**
- Q2.** (c) Write briefly on the role of lithology and geological structures in geomorphic landscape. **15**
-
- Q3.** (a) Discuss the scope and limitations of aerial photography. **20**
- Q3.** (b) Briefly enumerate the applications of GIS in various fields. **15**
- Q3.** (c) What are different strength parameters evaluated for rocks in engineering construction ? **15**
-
- Q4.** (a) What are the salient igneous rocks associated with mid oceanic ridges and trenches ? **20**
- Q4.** (b) Discuss the concept of isostasy. **15**
- Q4.** (c) Give a brief classification of meteorites and state their use in deciphering earth's interior. **15**
-

**SECTION – B**

- Q5.** Write short notes on the following in about **150** words **each**. **(5×10=50)**
- Q5.** (a) Micro fossils and their application in petroleum exploration. **10**
- Q5.** (b) Methods of stratigraphic correlation. **10**
- Q5.** (c) Hydrologic cycle. **10**
- Q5.** (d) Geological investigations for gravity dams. **10**
- Q5.** (e) Index fossils and their application in geological studies. **10**
-
- Q6.** (a) Discuss various flora and fauna of Gondwana sequence and write on their significance. **20**
- Q6.** (b) Write briefly on evolution of Siwalik basin and how the faunal assemblage help in deciphering the paleoclimatic conditions. **15**
- Q6.** (c) What are the major tectonic divisions and evolution of Himalayas during geological history ? **15**
-
- Q7.** (a) Discuss various aspects of porosity and permeability in rocks and give in detail various types of aquifers. **20**
- Q7.** (b) What are morphometric parameters used in basin analysis ? **15**
- Q7.** (c) Discuss various geological aspects of landslides and preventive measures for landslides. **15**
-
- Q8.** (a) Write on the evolutionary trends of suture line in cephalopoda and add note on their significance in Indian stratigraphy. **20**
- Q8.** (b) Enumerate the distribution of Dinosaurs in Indian sub-continent and state the causes of their extinction. **15**
- Q8.** (c) State briefly the methods of collection and preservation of microfossils. **15**
-

AL *